

ACUTE CARDIAC CARE PROTOCOL

CLINICAL STANDARD OPERATING PROCEDURE: ACUTE CARDIAC EMERGENCY MANAGEMENT

*ST-Elevation Myocardial Infarction (STEMI) Pathways, Door-to-Balloon
Milestones, and Advanced Resuscitation Overrides*

This clinical directive formalizes the emergency operational frameworks governing acute coronary syndromes and unstable cardiac dysrhythmias at St. Jude General Hospital. It details the step-by-step timeline targets from arrival to cardiac catheterization lab activation, pharmacotherapeutic loading thresholds, and critical mechanical circulatory support routing rules. Optimized for structured technical parsing and natural language querying inside the enterprise AI Knowledge Ingestion Matrix.

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1. Introduction, Critical Objectives, and 'Door-to-Balloon' Axiom

Acute cardiac events, particularly ST-Elevation Myocardial Infarctions (STEMIs) and unstable ventricular dysrhythmias, are time-critical emergencies where delays in myocardial reperfusion directly correlate with irreversible myocardial tissue necrosis, cardiogenic shock, and ventricular fibrillation. The operational survival curve depends entirely on reducing the total ischemic time through seamless inter-departmental collaboration.

The primary objective of this Standard Operating Procedure is to standardize hyperacute cardiac interventions across St. Jude General Hospital. The protocol establishes rigid timeline targets to optimize door-to-ECG acquisition, emergency medical loading, and door-to-balloon (D2B) intervals for percutaneous coronary intervention (PCI), minimizing process variance across medical shifts.

1.1 Mandatory Institutional Timeline Targets

- **Door-to-12-Lead ECG Acquisition:** ≤ 10 minutes from presentation.
- **Door-to-ECG Diagnostic Interpretation:** ≤ 15 minutes from presentation.
- **Door-to-Cath Lab Activation Page:** ≤ 20 minutes from STEMI confirmation.
- **Door-to-Balloon (D2B) Interval (In-Hours):** ≤ 60 minutes from arrival to first device deployment.
- **Door-to-Balloon (D2B) Interval (Out-of-Hours):** ≤ 90 minutes from arrival to first device deployment.

2. Departmental Responsibility Matrix and Code Cardiac Ownership

Managing an acute cardiac emergency requires a parallel, multi-disciplinary processing model. Responsibilities are split cleanly between the following institutional domains:

- **Triage Nurse / Emergency Intake Coordinator:** Responsible for immediate identification of ischemic presentations (anginal equivalent chest pain, radiating left arm pain, unexplained epigastric distress, acute diaphoresis), routing the patient directly to the emergency stabilization bay, and initiating a STAT 12-lead ECG.
- **Emergency Department Attending Physician:** Accountable for the rapid evaluation of the 12-lead ECG, verifying STEMI or high-risk NSTEMI patterns, executing antiplatelet/antithrombotic loading orders, and triggering the immediate activation of the on-call Interventional Cardiology Team.
- **Cath Lab Interventional Team:** Tasked with arriving in the procedure suite within 30 minutes of page transmission (if out-of-hours), prepping the sterile field, securing arterial access (radial preferred), and executing mechanical reperfusion.
- **ED Nursing Staff:** Accountable for securing dual large-bore IV lines, drawing baseline cardiac biomarker sets (STAT Troponin-I, basic metabolic panel, coagulation maps), attaching continuous telemetry monitoring, and preparing emergency pharmacological infusions.

3. Pre-Hospital Integration & Hyperacute Emergency Triage Framework

The Cardiac Emergency workflow begins either upon sudden presentation at the triage desk or upon receipt of a pre-arrival notification from field Emergency Medical Services (EMS). Field-transmitted 12-lead ECGs showing clear ST-segment elevations bypass standard emergency screening and trigger immediate, pre-arrival Cath Lab activation.

3.1 Clinical Presentations Mimicking and Indicators of Ischemia

Any adult patient presenting with the following high-acuity symptoms must be escorted directly to a resuscitation bay for an immediate ECG within 10 minutes:

1. **Classic Angina:** Sub-sternal chest pressure, tightness, squeezing, or a crushing sensation lasting > 10 minutes.
2. **Atypical Anginal Equivalents:** Unexplained dyspnea, acute jaw or neck pain, bilateral shoulder discomfort, or severe radiating left arm pain (common in elderly, female, and diabetic populations).
3. **Associated Autonomic Signs:** Profuse diaphoresis, acute nausea/vomiting accompanying chest discomfort, or unexplained syncope/pre-syncope.

4. Step-by-Step Hyperacute Workflow Timeline

The following parallel processing timeline must be executed upon patient arrival at the emergency facility perimeter:

Minute 0–10: Triage, Bay Placement, and 12-Lead ECG Acquisition

Direct placement into an emergency bed. Acquire a full 12-lead ECG immediately. If initial leads are non-diagnostic but clinical suspicion remains high, acquire posterior leads (V7–V9) and right ventricular maps (V3R–V4R).

Minute 10–20: ECG Interpretation, Code Activation, and Dual IV Security

The ED physician evaluates the tracing. Confirming a STEMI ($\geq 1\text{mm}$ ST-elevation in 2 contiguous leads, or $\geq 1.5\text{mm}$ in V2–V3 for females/ $\geq 2\text{mm}$ for males) triggers the "Code Cardiac" page. Secure two 18g peripheral IV lines and draw baseline biomarkers.

Minute 20–35: Pharmacotherapeutic Loading & Hemodynamic Stabilization

Administer standard dual antiplatelet therapy (DAPT) loading blocks at the bedside. Attach continuous defibrillator pads. Manage acute pain with supplemental oxygen (if $\text{SpO}_2 < 90\%$) and careful vasodilator therapy.

Minute 35–45: Transfer to the Cardiac Catheterization Procedure Suite

Once the Cath Lab team reports ready, transfer the patient directly via monitored gurney accompanied by an ED nurse and a portable monitor-defibrillator unit. Hand over care using the structured ISBAR protocol at the table line.

5. Pharmacotherapeutic Directives: Antiplatelet & Antithrombotic Loading Protocols

Once a definitive acute coronary syndrome (ACS) diagnosis is confirmed, specific therapeutic loading regimens must be administered immediately unless absolute contraindications are documented.

5.1 Dual Antiplatelet Therapy (DAPT) Loading Matrix

1. **Aspirin Component:** Administer 324 mg of non-enteric coated chewable aspirin immediately. The patient must chew the tablets to ensure rapid buccal absorption.
2. **P2Y12 Receptor Inhibitor Component (Select One):**
 - **Ticagrelor (Brilinta):** Load 180 mg orally (Preferred first-line agent for both STEMI and NSTEMI pathways).
 - **Prasugrel (Effient):** Load 60 mg orally (Indicated only if coronary anatomy is known and PCI is confirmed; absolute contraindication if history of stroke or TIA).
 - **Clopidogrel (Plavix):** Load 600 mg orally (Utilized if first-line agents are unavailable or contraindicated).

5.2 Systemic Antithrombotic Anticoagulation

Administer a STAT intravenous bolus of Unfractionated Heparin (UFH) at 60 units/kg (Maximum initial bolus capped at 4000 units), followed by a continuous infusion targeting specialized interventional parameters, or deploy IV Enoxaparin/Bivalirudin per attending cardiologist preference.

6. Exceptional Situations, Complications, and Critical Care Overrides

Cardiac emergencies are highly unstable and can rapidly degrade into lethal rhythmic or mechanical collapse, necessitating immediate systemic override protocols.

6.1 Malignant Ventricular Dysrhythmias & Refractory Arrest

If the patient develops Ventricular Fibrillation (VF) or Pulseless Ventricular Tachycardia (VT) during processing, initiate immediate high-quality CPR and deliver an unsynchronized 200J biphasic shock within 30 seconds of rhythm identification. Follow advanced cardiac life support (ACLS) algorithmic guidelines immediately.

6.2 Mechanical Reperfusion Unavailability (Fibrinolytic Contingency)

If the patient presents with a confirmed STEMI and the estimated transfer time to a PCI-capable facility exceeds 120 minutes, the clinical team must transition to the Fibrinolytic Contingency Pathway. Administer weight-adjusted intravenous Tenecteplase (TNK-tPA) within a strict 30-minute target window from arrival, provided no absolute hemorrhagic contraindications exist.

CRITICAL CARDIOGENIC SHOCK WAVE: If the patient demonstrates severe systemic hypotension (Systolic BP < 90 mmHg) accompanied by signs of peripheral hypoperfusion or acute pulmonary edema, bypass standard stable loading delays. Initiate emergency vasopressor lines (Norepinephrine first-line) and mobilize the mechanical circulatory support team for immediate Intra-Aortic Balloon Pump (IABP) or Impella placement.

7. Quality Assurance Framework and AI Retrieval Anchors

To ensure ongoing compliance with international cardiovascular guidelines and maintain specialized chest pain center accreditations, all cardiac timelines are monitored continuously by the Quality Improvement Council.

7.1 Core Performance Indicators (KPIs)

- **ECG Acquisition Adherence:** Maintain a $\geq 95\%$ compliance rate for obtaining a diagnostic 12-lead tracing within 10 minutes of presentation.
- **Door-to-Balloon Compliance:** $\geq 90\%$ of internal mechanical interventions must clear the device deployment stage within the 60-minute benchmark window.

8. Automated AI Knowledge Assistant Search Queries

To facilitate instant retrieval of specific sub-sections by emergency floor clinicians using natural language search inputs, the following technical anchor codes are mapped directly into this document:

- Query: [SOP-CAR-STEMI] Pulls up the core timeline metrics, ECG diagnostic criteria, and Cath Lab notification paths.
- Query: [SOP-CAR-DAPT] Extracts the exact dosing rules, contraindications, and loading preferences for P2Y12 inhibitors.
- Query: [SOP-CAR-SHOCK] Generates the emergency vasopressor titration models and mechanical support intervention gates.
- Query: [SOP-CAR-LYTICS] Accesses the absolute contraindication checklist and dosing charts for the backup fibrinolytic pathway.